Software Engineering Lab-1
Name – Avaneesh Aroor
SRN – PES2UG24CS098
Section - 5B

Lab 1: Requirements Table

Problem Statement: Micro-Lending & Peer Credit Risk Assessor, FinTech & Financial Inclusion

Scenario: A peer to peer micro-lending platform serves borrowers who have little or no formal credit history. Borrowers submit small loan requests, an automated assessor grades each request for credit risk using alternative data, individual peer lenders browse the marketplace and fund requests in small increments under diversification limits, and the platform tracks repayments and flags delinquency. Stakeholders/Actors: Borrower, Peer Lender, Credit Risk Analyst.

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enable a borrower to create a loan request containing the requested principal amount, loan purpose, repayment tenure, and a verified KYC identity document with a linked bank account.	High	Pass: The loan request is published to the lending marketplace only after the KYC document and bank account are successfully validated. Fail: A request with an unverified or malformed KYC document is published.	Establishes the core credit record that the scoring engine assesses and that peer lenders fund; without verified identity and bank linkage, borrower authenticity and repayment capacity cannot be confirmed.
FR-002	Functional	The system shall compute a credit risk score and an associated risk grade (A to E) for every loan request from alternative data (past repayment history, bank cash flow patterns, and peer endorsements), and attach the grade to the listing before publication.	High	Pass: Every published listing displays a risk grade derived from all three data sources, with the contributing factors recorded in an audit log. Fail: A listing is published with a missing risk grade or a grade computed from incomplete input data.	Gives thin file borrowers a fair assessment path while giving lenders a standardised, auditable basis for pricing risk instead of relying on subjective judgement.
FR-003	Functional	The system shall allow a verified peer lender to pledge funds toward an open loan request in increments, while restricting any single lender to no more than 20% of the principal of that request.	High	Pass: A pledge that would take a lender above 20% of the request principal is rejected with an on screen message, and the remaining amount updates immediately. Fail: The system accepts a pledge that breaches the 20% concentration cap or over funds the request.	Enforces portfolio diversification so that no single lender carries excessive exposure to one borrower, which is central to keeping peer lending losses survivable.
FR-004	Functional	The system shall verify the borrower's identity via a one time password (OTP) sent to the registered email address or phone number before any funds are disbursed from a fully funded loan request.	High	Pass: Disbursal proceeds only after the correct OTP is entered within a 5 minute validity window. Fail: Funds are disbursed without OTP verification, or an expired OTP is accepted.	Core anti fraud control that prevents account takeover and synthetic identity borrowers from redirecting disbursed funds at the final and most damaging step.
FR-005	Functional	The system shall generate an amortised repayment schedule for each disbursed loan and update the loan status (current, late, or default) as instalments fall due, notifying both borrower and lenders of any change.	Medium	Pass: The schedule reflects a recorded repayment and any resulting status change within the NFR-001 latency target, and notifications are dispatched to all affected parties. Fail: Stale loan status or instalment balances are shown after a repayment has been recorded.	Gives borrowers and lenders a transparent, trustworthy view of outstanding obligations and portfolio health throughout the life of the loan.
NFR-001	Nonfunctional (Performance & Security)	The credit risk scoring and marketplace listing calculations shall update in real time with an API response latency under 300 ms.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load.	Lenders make funding decisions against a moving marketplace; noticeable lag in scores or remaining amounts causes over pledging and undermines confidence in the platform.

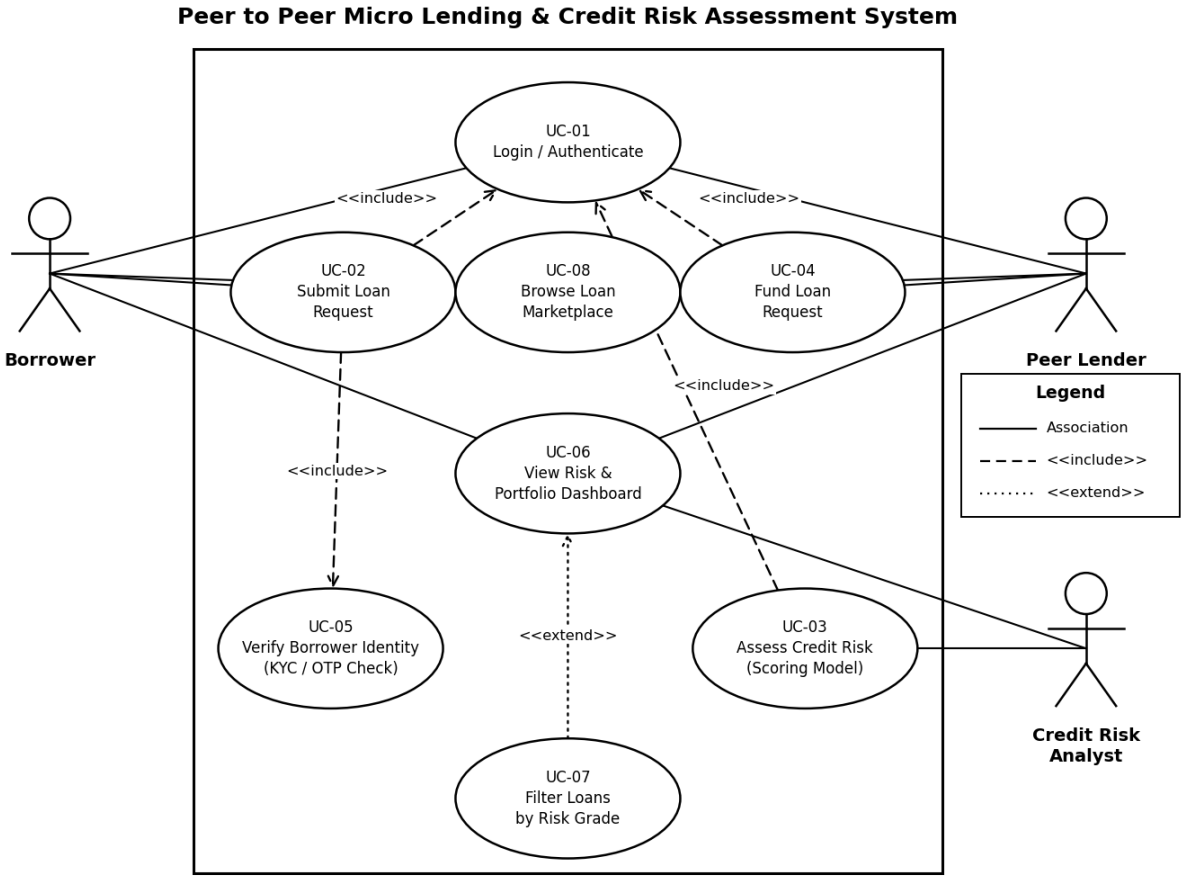
Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
NFR-002	Nonfunctional (Security)	The system shall reject fraudulent or duplicate loan applications (reused identity documents, repeated device or IP fingerprints, disposable email addresses) with at least 99.5% accuracy under simulated adversarial load, while blocking no more than 1% of legitimate applicants.	High	Pass: A security test suite combining a simulated fraud ring with a legitimate applicant sample confirms 99.5% or higher malicious application rejection and a 1% or lower false positive rate. Fail: Either threshold is missed.	Protects lender capital from organised fraud while ensuring genuine underserved borrowers, who are the platform's purpose, are not unfairly excluded.

Note: FR-001 and NFR-001 are the instructor provided baseline requirements (description and acceptance criteria reproduced as given); FR-002 to FR-005 and NFR-002 were elicited to complete the required set of 5 FRs and 2 NFRs for this scenario.

Lab 1: UML Use Case Diagram

Problem Statement: Micro-Lending & Peer Credit Risk Assessor

Actors: Borrower, Peer Lender, Credit Risk Analyst. Includes at least one <<include>> and one <<extend>> relationship, as required.



Lab 1: Use Case Flow Specification

Problem Statement: Micro-Lending & Peer Credit Risk Assessor

Use Case: UC-02 Submit Loan Request

Primary Actor: Borrower

Includes: UC-05 Verify Borrower Identity (KYC / OTP Check)

Preconditions

The lending marketplace is open for new submissions, the borrower holds a registered platform account with a linked bank account, and the borrower has internet access. The borrower has no other loan request currently in an unfunded or active state.

Trigger

The borrower selects "Request a Loan" from the borrower dashboard.

Main Success Scenario

1. Borrower opens the borrower dashboard and selects "Request a Loan."

2. System displays the loan request form, since this borrower has no active or pending request.

3. Borrower enters the principal amount, loan purpose, and repayment tenure, and uploads a KYC identity document.

4. System checks whether the borrower holds a verified KYC session; if not, the Verify Borrower Identity use case (UC-05) executes via «include»: the system sends a one time password (OTP) to the borrower's registered email or phone.

5. Borrower enters the OTP within the 5 minute validity window.

6. System validates the OTP, matches the uploaded document against the registered identity, and marks the borrower's session as verified.

7. System invokes the credit risk assessor, which computes a risk score and assigns a risk grade (A to E) from repayment history, bank cash flow patterns, and peer endorsements.

8. System publishes the loan request to the lending marketplace with its risk grade and updates the marketplace listing in real time (see NFR-001).

9. System displays a confirmation message: "Your loan request has been published to the marketplace."

10. Use case ends successfully.

Alternate Flow

5a. Invalid or Expired OTP (Failed Identity Verification)

a. System displays "Incorrect or expired code, please try again."

b. Borrower may request a new OTP, up to 3 resend attempts within the submission session.

c. If the borrower enters a valid OTP within the allowed attempts, the flow resumes at main step 6.

d. If the borrower exhausts all 3 attempts without a valid OTP, the system blocks further OTP requests from that email/phone for 15 minutes, displays "Too many attempts, please try again later," saves the request as an unpublished draft, and the use case ends without a loan request being published.

7a. Risk Grade Below Platform Threshold

a. If the computed risk grade falls below the platform's minimum listing threshold, the system does not publish the request.

b. System displays the principal contributing risk factors and the earliest date on which the borrower may reapply.

c. The request is retained for review by the Credit Risk Analyst, and the use case ends without publication.

Postconditions

Success: Exactly one published loan request exists for the borrower, carrying a computed risk grade, and it is visible to peer lenders in the marketplace.

Failure: No loan request is published for the borrower; any uploaded data is retained as an unpublished draft, and the borrower's prior verification state, if any, is unaffected and they may retry within policy limits.